

Modeling and Synchronization Control of a Dual Drive Industrial Gantry Stage

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Abstract—Dual drive gantry mechanisms, in which two actuators are physically connected through a crossbeam to provide a joint thrust for higher acceleration, are widely used in high-speed or heavy payload precision motion control applications. Synchronizing motions of the two actuators and avoiding excessive internal forces are both essential to the smooth operation of these systems. This paper presents a novel synchronization control scheme with thrust allocation that achieves not only better motion synchronization of the two actuators but also simultaneous regulation of internal forces. With decreased internal forces, lesser wear in sliding parts, lower energy consumption, and higher performances can be achieved for dual drive gantry stages. In contrast, existing synchronization control approaches mainly focus on pure motion compensation and synchronization of the two actuators only. Due to the near-rigid physical coupling between the two actuators, excessive internal forces or the “pull and drag” phenomena could still exhibit under existing approaches even when their motions are quite well synchronized. Comparative experimental results are also obtained to show the superiority of the proposed synchronization controller over existing ones in practical applications.

Index Terms—Adaptive robust control (ARC), linear motor, motion control, redundant drive, system identification, synchronization control.

I. INTRODUCTION

MULTIPLE physically coupled actuators are often used to power the motion of a single-degree-of-freedom (DOF) axis when larger thrust and high precision motion are required.

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A typical example of such systems is the dual drive industrial gantry stage studied in this paper, in which two linear motors [1]–[9] mounted on two parallel guides and connected through a crossbeam are used in Y -axis in order to generate higher acceleration/speed movement of the X -axis of a cartesian robotic system [10]. This type of moving stages has been widely deployed in automation processes, such as printed circuit board manufacturing, precision metrology, and circuit assembly. Physical connection of the two motors through the crossbeam is necessary for joint thrust generation.

There have been a large amount of work done on both the synchronized motion control [11]–[15] and the coordinated motion control of multiple machine axes [16]–[21]. The objective of the former is to control various actuators or axes to follow the same motion, while the latter is to coordinate motion of various axes such that the overall motion of the end-effector/tool follows certain prespecified motions in task space (e.g., contour following [22]). Earlier synchronized motion control schemes can be classified into the following three categories [11].

- 1) Synchronized master motion command generation, in which each motor has its own individual control loop but receives the synchronized motion commands.
- 2) Master-slave motion control, in which the commands to the slave motor control loop are to follow the master motor motion.
- 3) Relative dynamic stiffness motion control [12], [13], in which a synchronization compensator is constructed based on the relative position and velocity of the two machine axes to be synchronized.

The first two are intuitive but have inherent performance limitations due to nonsharing of feedback information (e.g., feedback information of the slave axis is not shared with the master in the second category). The third category can be thought as a special case of the cross-coupled coordinated motion control [16] by simply defining the contour error to be the relative position of the two machine axes. As in the coordinated motion controls [16], [22], [23], latest synchronization motion controls mainly focus on the use of feedback information of all axes and/or more refined control design methodologies to achieve better motion synchronization under different drive characteristics and loading conditions of individual axis [14], [24]–[26].

It should be noted that there are subtle but fundamental differences between the synchronization control of dual motors studied in this paper and the coordinated motion control of two-axes machines for contour following tasks. Dynamics of individual axes in the synchronized control are completely coupled. This

is in contrast to the essentially decoupled axis dynamics of the coordinated motion controls. Specifically, due to the mechanical mechanism of coupling bridge used in the dual-motor drive system constructions, physical constraints do exist in the positions of the two motors to restrict their relative motion or physically synchronize their motions. This causes large internal force when motion control of the two motors are not well synchronized, leading to detrimental effects such as the “pull and drag” phenomena and the excessive wear in sliding parts [27], control chattering/saturation, and even damage of mechanical components. Appropriate control strategies that not only synchronize motions of the two motors but also avoid large internal forces are keys to the smooth operations of these systems. However, the problem of internal force regulation has been essentially ignored in existing synchronization control schemes. This might be caused by the complexity of having a good knowledge of the completely coupled dynamics of the two actuators due to the mechanical coupling bridge. Specifically, recognizing the potential detrimental effects of rigid mechanical coupling when controls of two motors are not well synchronized, compliance mechanisms are actively introduced in [27] to avoid excessive wear in sliding parts and to provide certain hardware safety. However, doing so compromises the mechanical accuracy that can be achieved by these dual drive systems and, thus, defeats the main hardware benefit of using these H-type gantry systems. Lagrangian equations are also used to model the gantry systems in [28] by choosing the generalized coordinates to be the displacements of the two motors and the rotational angle of the crossbar. However, such handling of motion constraints directly contradicts with the rigid crossbar connection of the two motors implicitly assumed in the paper. Namely, with a rigid crossbar connection assumption, mathematically, a holonomic kinematic constraint exists between motions of the two motors and the overall system should have only one DOF when using Lagrangian equations to obtain the system dynamics, not the three DOFs assumed in this paper. Later on, a lumped-parameter model is presented [25] by treating the connections between the crossbar and the two parallel linear guides of the two motors as pure rotational flexible joints with lumped rotational spring stiffness. Again, such a modeling directly contradicts with the rigid crossbar assumption made in this paper. Namely, when the rigid crossbar has a rotational motion as assumed in the paper, there must be lateral displacements between the bar and the two fixed parallel linear guides along the Y -axis. Consequently, the connections between the bar and the two linear guides cannot be assumed to be pure rotational flexible joints.

In this paper, a novel physics-based lumped parameter modeling approach will be taken to properly address the complex dynamic coupling issues as already mentioned. Though not rigorous and complete, such a modeling approach has been touched on partly in our conference publication [10]. Specifically, recognizing the fact that rotations of the crossbar are actually caused by the linear spring stiffness effect of the ball bearings between the rotors and the linear guides of the two motors, a dynamic model based on lumped bearing stiffness will be derived first. The model will then be validated with appropriate system identification experiments.

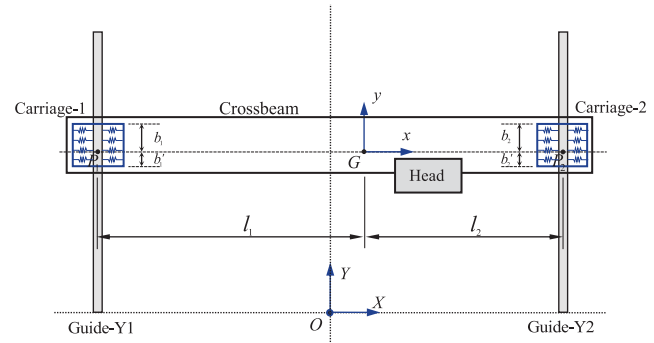


Fig. 1. Dual drive gantry with ball bearings at two parallel linear guides.

As opposed to the pure motion synchronization of existing controls, this paper also focuses on simultaneous regulation of internal forces to achieve smooth operation of these dual drive systems. Specifically, with the knowledge of the interaction between the crossbar and the linear guides and the resulting rotational dynamics, a novel synchronization controller will be presented in which thrust allocation techniques are used to avoid excessive internal forces. *For comparison, as a typical example of existing pure motion synchronization controls, a synchronization controller with cross-coupled compensation will be presented as well.* The adaptive robust control (ARC) strategy is used in both synchronization control methods to synthesize specific control algorithms to address the issues of significant parametric uncertainties and uncertain nonlinearities in practical systems. Comparative experiments have been performed on a dual drive industrial gantry stage and the results validate the effectiveness and superiority of the proposed synchronization controller over existing ones in practical applications. Specifically, with decreased internal forces achieved by the proposed thrust allocation based synchronization strategy, lesser wear in sliding parts, lower energy consumption, and higher performances can be achieved for dual drive gantry stages. In contrast, existing synchronization control approaches mainly focus on pure motion compensation and synchronization of the two actuators only. Due to the near-rigid physical coupling between the two actuators, excessive internal forces or the “pull and drag” phenomena could still exhibit under existing approaches even when their motions are quite well synchronized.

II. PHYSICAL MODELING

A. Configuration and Kinematics

The schematic vertical view of a dual drive gantry is shown in Fig. 1, in which two linear guides Y1 and Y2 are aligned parallel to each other and a crossbeam is mounted across them. The crossbeam is driven along the direction of the linear guides by two linear motors Y1 and Y2 with their rotors rigidly attached to the crossbeam at the two ends while their stators mounted on the two linear guides, respectively. The gantry stage head is either rigidly attached to the crossbeam or driven along the crossbeam by another linear motor for an additional X -axis directional motion. As this paper focuses on the synchronized control of the two motors Y1 and Y2 for Y -axis motion of the head only,

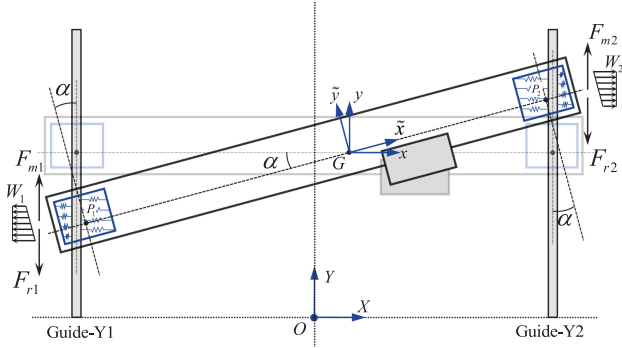


Fig. 2. Motion and dynamics of the rigid moving body.

for simplicity, it is assumed that the head is rigidly attached to the crossbeam.

Because of many advantages, such as high stiffness, good reliability, and low cost, linear motion guides using recirculating ball bearing [29] are widely used in precise positioning systems. The carriage blocks of the two linear guides are rigidly fixed to the crossbeam as represented by carriage-1 and carriage-2 in Fig. 1. The bearing balls between the carriage blocks and the guide rails are normally considered as elastic due to their relatively poorer rigidity. Thus, in practice, in addition to their intended longitudinal motions along the guide rails, small lateral motions of the carriage blocks exist when significant lateral forces are applied to the carriage blocks as well, which is indeed the scenario in this paper. Furthermore, due to precompression of the bearing balls by preloads during installation, the elastic effect of the bearing balls on such lateral motions of the carriage blocks can be modeled by distributed horizontal two-way springs. It is assumed that the dual drive gantry mechanism is manufactured and installed perfectly, i.e., the geometric parallelism of the two linear guides is sufficiently guaranteed and the preloads of the bearings are appropriately applied such that, when the crossbeam is perpendicular to the parallel guide rails, no net lateral forces are applied to each of the carriage blocks by their bearing balls or the distributed horizontal two-way springs are perfectly balanced as illustrated in Fig. 1.

Let OXY be the fixed inertia coordinate frame with origin at the midpoint between the two linear guides and Y -axis parallel to the linear guides. Let G be the mass center of the entire rigid moving body including the crossbeam, the head, the carriage blocks, and the rotors of the two linear motors Y1 and Y2. Let Gxy be a moving coordinate frame at G that parallels with OXY and $G\tilde{x}\tilde{y}$ a body-fixed coordinate frame to the crossbeam at G with its $\tilde{x}$ -axis paralleling the longitudinal direction of the crossbeam as shown in Fig. 2. Denote the distances from the mass center G to the two linear guides as l_1 and l_2 , respectively, i.e., $|GP_1| = l_1$ and $|GP_2| = l_2$, in which P_1 and P_2 are the two points on the crossbeam representing the intersections of the longitudinal direction line passing through G with the two linear guides when $G\tilde{x}\tilde{y}$ is at Gxy as in Fig. 1. Denote the active lengths of the carriage block (i.e., the lengths of the distributed load zone of the ball bearings along y -axis direction seen in Fig. 1) of Guide Y1 before and after P_1 as b_1 and b'_1 , and of Guide Y2 before and after P_2 as b_2 and b'_2 , respectively.

With the above definitions, the planar motion of the crossbeam can be completely described by the following three generalized coordinates: two position coordinates of G , x_G and y_G , and the rotation angle of the body-fixed coordinate frame $G\tilde{x}\tilde{y}$, α , in the world frame OXY as shown in Fig. 2. The positions of P_1 and P_2 under this planar rigid body motion are denoted as $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ in OXY , respectively, in which y_1 and y_2 are measured through linear encoders mounted on the two guides and are assumed to be available for controller design in the following. Thus, as seen from Fig. 2,¹

$$\begin{aligned} x_1 &= x_G - l_1 \cos(\alpha) \approx x_G - l_1 \\ y_1 &= y_G - l_1 \sin(\alpha) \approx y_G - l_1 \alpha \\ x_2 &= x_G + l_2 \cos(\alpha) \approx x_G + l_2 \\ y_2 &= y_G + l_2 \sin(\alpha) \approx y_G + l_2 \alpha \end{aligned} \quad (1)$$

from which the real-time feedback of y_G and α can be obtained by

$$\begin{aligned} y_G &= \frac{l_2}{l} y_1 + \frac{l_1}{l} y_2 \\ \alpha &= \frac{1}{l} (y_2 - y_1) \end{aligned} \quad (2)$$

where $l = l_1 + l_2$.

B. Dynamics

As shown in Fig. 2, the electromagnetic driving forces produced by the two linear motors are F_{m1} and F_{m2} , and the frictions of the two linear guides are F_{r1} and F_{r2} , respectively. Under the assumption that the current-loops of the motor drive amplifiers are fast enough to be negligible, these forces can be modeled in the same way as in the control of single linear motors [1] as

$$\begin{aligned} F_{m1} &= K_1 u_1, & F_{r1} &= B_1 \dot{y}_1 + A_1 S_f(\dot{y}_1) \\ F_{m2} &= K_2 u_2, & F_{r2} &= B_2 \dot{y}_2 + A_2 S_f(\dot{y}_2) \end{aligned} \quad (3)$$

where u_1 and u_2 are the current control input commands to the two motors with K_1 and K_2 being the corresponding force constants, B_1 and B_2 are the viscous friction coefficients, A_1 and A_2 are the Coulomb friction coefficients of the two linear guides, respectively, in which $S_f(\cdot)$ is a continuous function chosen to approximate the sign function $\text{sgn}(\cdot)$ used in traditional Coulomb friction modeling for realistic model compensation designs. Different from the control of individual linear motors [1], rotational motion of the crossbeam in the synchronized controls of dual motors also leads to the unbalanced lateral deformations of the ball bearings and, consequently, the distributed lateral forces to the carriage blocks W_1 and W_2 shown in Fig. 2, which can be obtained as follows.

From (1), the lateral deformation of ball bearings along X -axis at P_1 and P_2 are $s_{P_1} = x_1 + 0.5l \approx \tilde{x}_G$ and $s_{P_2} = x_2 - 0.5l \approx \tilde{x}_G$, respectively, in which $\tilde{x}_G = x_G - 0.5(l_1 - l_2)$ represents

¹ Throughout the paper, small rotational angle α is assumed and any nonlinear functions of α will be approximated by their linear terms only.

the displacement of G with respect to its equilibrium position along X -axis. Let ζ_i be the geometric distance to P_i of guide Y_i , $i = 1, 2$, along $\tilde{y}$ -axis. Then, the deformations of the springs at each distributed point at $\zeta_i \in [-b'_i, b_i]$, $i = 1, 2$, can be described by

$$s_{\zeta_i} = s_{P_i} - \zeta_i \sin(\alpha) \approx \tilde{x}_G - \zeta_i \alpha \quad (4)$$

and the distributed forces w_i of unit length are

$$w_i(\zeta_i) = -k_{si}s_{\zeta_i} \approx -k_{si}(\tilde{x}_G - \zeta_i \alpha) \quad (5)$$

where k_{si} represents the equivalent stiffness of the horizontal springs of guide Y_i per unit distributed length. The dynamics of the rigid moving body are thus described by

$$M\ddot{\tilde{x}}_G = \sum_{i=1}^2 \int_{-b'_i}^{b_i} w_i(\zeta_i) d\zeta_i \approx -K_s \tilde{x}_G + C_{x\alpha} \alpha \quad (6)$$

$$M\ddot{y}_G = F_{m1} + F_{m2} - F_{r1} - F_{r2} \quad (7)$$

$$J\ddot{\alpha} = -(F_{m1} - F_{r1})l_1 + (F_{m2} - F_{r2})l_2 + \sum_{i=1}^2 M_{\alpha i} \quad (8)$$

where M and J represent, respectively, the mass inertia and the rotational inertia of the entire rigid body, $K_s = \sum_{i=1}^2 k_{si}(b_i + b'_i)$, $C_{x\alpha} = \sum_{i=1}^2 k_{si}(b_i^2 - b_i'^2)/2$, and $M_{\alpha i}$ are the moments generated by the distributed lateral forces of two guides, which can be calculated as

$$\begin{aligned} \sum_{i=1}^2 M_{\alpha i} &= - \int_{-b'_1}^{b_1} w_1(\zeta_1)[-l_1 \sin(\alpha) + \zeta_1 \cos(\alpha)] d\zeta_1 \\ &\quad - \int_{-b'_2}^{b_2} w_2(\zeta_2)[l_2 \sin(\alpha) + \zeta_2 \cos(\alpha)] d\zeta_2 \\ &\approx C_{x\alpha} \tilde{x}_G - K_{\alpha} \alpha \end{aligned} \quad (9)$$

in which $K_{\alpha} = \sum_{i=1}^2 \frac{1}{3} k_{si}(b_i^3 + b_i'^3)$. It is thus easy to see that, due to the large equivalent bearing stiffness k_{si} , both the resulting lateral displacement stiffness K_s in (6) and the rotational stiffness K_{α} in (9) are large, leading to small rotational angles and lateral displacements in practice.

C. High-Order Linear Dynamics

To obtain the underlying high-order linear dynamics, we temporarily ignore nonlinear Coulomb frictions in this subsection. Also, in most applications, the coupling coefficient $C_{x\alpha}$ between the lateral displacement dynamics (6) and the rotational angle dynamics (8) is negligible due to almost symmetry construction of the moving rigid-body, which is assumed in this paper. Under this assumption, the lateral dynamics (6) are completely decoupled from the remaining dynamics, and thus will be neglected in the rest of the paper, i.e., assuming $\tilde{x}_G \approx 0$. Noting (1) and (3), the remaining dynamics are thus simplified to

$$M\ddot{y}_G = K_1 u_1 + K_2 u_2 - B_t \dot{y}_G + C_{\dot{y}\dot{\alpha}} \dot{\alpha} \quad (10)$$

$$J\ddot{\alpha} = -K_1 l_1 u_1 + K_2 l_2 u_2 + C_{\dot{y}\dot{\alpha}} \dot{y}_G - B_{\alpha} \dot{\alpha} - K_{\alpha} \alpha \quad (11)$$

where $B_t = B_1 + B_2$, $C_{\dot{y}\dot{\alpha}} = B_1 l_1 - B_2 l_2$, and $B_{\alpha} = B_1 l_1^2 + B_2 l_2^2$. The transfer functions from input u_1 to y_G and from u_2 to y_G can subsequently be obtained as

$$P_{G1}(s) = \frac{K_1(Js^2 + B_2 l_2 s + K_{\alpha})}{D(s)} \quad (12)$$

$$P_{G2}(s) = \frac{K_2(Js^2 + B_1 l_1 s + K_{\alpha})}{D(s)} \quad (13)$$

where the denominator denoted by $D(s)$ is given by

$$\begin{aligned} D(s) &= JM s^4 + (JB_t + MB_{\alpha}) s^3 \\ &\quad + (B_t B_{\alpha} + K_{\alpha} M - C_{\dot{y}\dot{\alpha}}^2) s^2 + K_{\alpha} B_t s. \end{aligned} \quad (14)$$

Noting $K_{\alpha} \gg C_{\dot{y}\dot{\alpha}}^2/M$ in practice, $K_{\alpha} M - C_{\dot{y}\dot{\alpha}}^2 \approx K_{\alpha} M$, and the transfer functions $P_{G1}(s)$ and $P_{G2}(s)$ can be decomposed to

$$P_{G1}(s) = \frac{K_1}{Ms^2 + B_t s} \frac{Js^2 + B_2 l_2 s + K_{\alpha}}{Js^2 + B_{\alpha} s + K_{\alpha}} \quad (15)$$

$$P_{G2}(s) = \frac{K_2}{Ms^2 + B_t s} \frac{Js^2 + B_1 l_1 s + K_{\alpha}}{Js^2 + B_{\alpha} s + K_{\alpha}}. \quad (16)$$

Similarly, the transfer functions from input u_1 and u_2 to α can be obtained as

$$P_{\alpha 1}(s) = \frac{-K_1(Ml_1 s + B_2 l)}{Ms + B_t} \frac{1}{Js^2 + B_{\alpha} s + K_{\alpha}} \quad (17)$$

$$P_{\alpha 2}(s) = \frac{K_2(Ml_2 s + B_1 l)}{Ms + B_t} \frac{1}{Js^2 + B_{\alpha} s + K_{\alpha}}. \quad (18)$$

It is thus clear that pure rigid-body dynamics of the system are described by $Ms + B_t$ and the linear spring stiffness effects of the ball bearings are captured by $Js^2 + B_{\alpha} s + K_{\alpha}$ in the denominators of all the transfer functions; the flexible mode has an undamped resonant frequency of $\sqrt{\frac{K_{\alpha}}{J}}$. In terms of directly measured signals, the transfer functions from input u_1 and u_2 to y_1 and y_2 , respectively, can be obtained as

$$P_{11n}(s) = \frac{K_1}{Ms^2 + B_t s} \frac{(J + Ml_1^2) s^2 + B_2 l^2 s + K_{\alpha}}{Js^2 + B_{\alpha} s + K_{\alpha}} \quad (19)$$

$$P_{12n}(s) = \frac{K_2}{Ms^2 + B_t s} \frac{(J - Ml_1 l_2) s^2 + K_{\alpha}}{Js^2 + B_{\alpha} s + K_{\alpha}} \quad (20)$$

$$P_{21n}(s) = \frac{K_1}{Ms^2 + B_t s} \frac{(J - Ml_1 l_2) s^2 + K_{\alpha}}{Js^2 + B_{\alpha} s + K_{\alpha}} \quad (21)$$

$$P_{22n}(s) = \frac{K_2}{Ms^2 + B_t s} \frac{(J + Ml_2^2) s^2 + B_1 l^2 s + K_{\alpha}}{Js^2 + B_{\alpha} s + K_{\alpha}} \quad (22)$$

which will be validated in the following identification experiments.

III. IDENTIFICATION EXPERIMENTS

A. Experiment Setup

The modified Anorad HER-510-510-AA1-C-CC2 gantry from Rockwell Automation detailed in [10] is used as a test-bed, in which the two linear motors are measured through linear encoders with a resolution of 0.5 μm after quadrature. The same

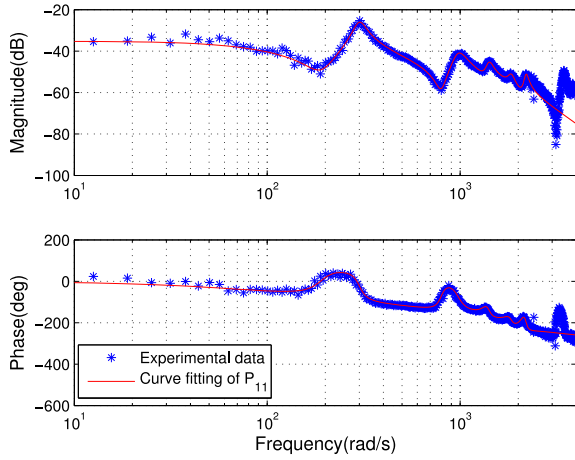


Fig. 3. Identification of $P_{11}(s)$ in frequency domain.

dSPACE CLP1103 controller board as in [21] is used to implement the proposed control algorithm in experiments, in which the velocity signal is obtained by the difference of two consecutive position measurements. The dSPACE controller board is directly programmable under the MATLAB Simulink block diagram environment, which is a versatile controller board commonly used in industry for demanding tasks in rapid control prototyping, with a PowerPC 750GX running at 933 MHz, a 32 MB application SDRAM as program memory, and a 96 MB communication SDRAM for data storage and exchange with the host PC. All implemented control algorithms are written as an S-function based on C codes at the host PC with MATLAB Simulink software. Using the propriety interface software provided by dSPACE company, real-time control codes are compiled directly from MATLAB Simulink control codes and downloaded to the dSPACE controller board, which executes the program with a sampling period of 0.0002 s.

B. Identification Results

Frequency response identification tests are carried out to reveal linear dynamics of the system to validate the proposed dynamic modeling.

1) *Identification of $P_{11}(s)$* : The excitation input signal, which is generated using the `idinput` command [`sinesum frequ`] = `idinput([10000 1 50], 'sine', [0 1], [-1 1], [4999 50 1])` in MATLAB, is fed to motor Y1 and the resulting experimental frequency responses from input voltage u_1 to velocity $\dot{y}_1$ are shown in blue dots in Fig. 3. Fitting the obtained frequency responses by the red line in Fig. 3, the following overall transfer

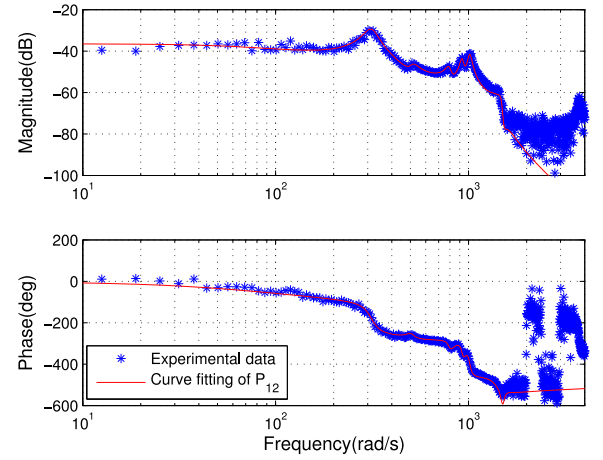


Fig. 4. Identification of $P_{12}(s)$ in frequency domain.

function is obtained

$$P_{11}(s) = G_{11a}(s)G_{11b}(s)G_{11c}(s) \quad (23)$$

in which

$$G_{11a}(s) = \frac{1}{0.64s + 57.6} \quad (24)$$

$$G_{11b}(s) = \frac{2.504(s^2 + 57.32s + 3.612 \times 10^4)}{s^2 + 42.97s + 9.046 \times 10^4} \quad (25)$$

equation (26) shown at the bottom of this page. It is seen from the plots that G_{11a} and G_{11b} match well with the proposed dynamics (19) that include the rigid-body dynamics, an antiresonant frequency around 190.1 rad/s, and a resonant frequency at about 300.8 rad/s. G_{11c} represents additional dynamics in much higher frequency range with some uncertainties. By comparing G_{11a} and G_{11b} with transfer function (19) in terms of antiresonant frequency and resonant frequency, $\frac{K_a}{J + Ml_1^2} = 3.612 \times 10^4$, $\frac{K_a}{J} = 9.046 \times 10^4$. These values are well in agreement with their theoretical values calculated based on specifications from the manufacturers that $M = 66.5$, $l_1 = 0.535$ and $l_2 = 0.505$, $J = 12.65$, and $K_a = 11.45 \times 10^5$.

2) *Identification of $P_{12}(s)$* : Feeding the excitation signal to motor Y2, the resulting experimental frequency responses from input voltage u_2 to velocity $\dot{y}_1$ are shown in Fig. 4. The curve-fitted transfer function whose frequency responses are shown in red lines is given by

$$P_{12}(s) = G_{12a}(s)G_{12b}(s)G_{12c}(s) \quad (27)$$

$$\begin{aligned} G_{11c}(s) = & \frac{1.283 \times 10^6 (s^2 + 87.17s + 6.369 \times 10^5) (s^2 + 200.4s + 1.824 \times 10^6)}{(s^2 + 160.9s + 9.542 \times 10^5) (s^2 + 130s + 1.986 \times 10^6)} \\ & \times \frac{(s^2 + 291.2s + 3.3 \times 10^6) (s^2 + 225.2s + 4.471 \times 10^6)}{(s^2 + 160.8s + 3.387 \times 10^6) (s^2 + 120.4s + 4.808 \times 10^6)} \\ & \times \frac{s + 1.465 \times 10^4}{(s^2 + 2454s + 1.911 \times 10^6) (s + 5464)}. \end{aligned} \quad (26)$$

where

$$G_{12a}(s) = \frac{0.82}{0.64s + 57.6} \quad (28)$$

$$G_{12b}(s) = \frac{-0.4355(s + 475.5)(s - 475.5)}{s^2 + 52.97s + 9.846 \times 10^4} \quad (29)$$

equation (30) shown at the bottom of this page. Again, the identified G_{12a} and G_{12b} match well with the proposed dynamics (20). From the specification of the motors that $K_1 = 74.2 \text{ N/V}$ and $K_2 = 55.6 \text{ N/V}$, we have the calculated value of $\frac{K_2}{K_1} = 0.75$, which is in agreement with its identified value of $\frac{K_2}{K_1} = 0.82$ based on G_{11a} in (24) and G_{12a} in (28). The resonance dynamics due to the ball bearings in G_{12b} also agree well with G_{11b} , with the resonant frequency in G_{12b} of 313.8 rad/s staying quite close to 300.8 rad/s of G_{11b} . In addition, with $J = 12.65$ and $K_\alpha = 11.45 \times 10^5$, $\frac{K_\alpha}{J-MI_{1/2}} = -2.154 \times 10^5$. Thus, noting (20), one can predict that there exists a non-minimum-phase zero at 464.1 rad/s , well in agreement with the identified results of 475.5 rad/s based on the numerator of G_{12b} .

Identification results of $P_{21}(s)$ and $P_{22}(s)$ have been obtained in a similar way but omitted here due to space limit. Again, those experimental results agree well with the derived dynamics and the identification results as already shown. It is thus safe to conclude from these experimental identification results that the dominant linear dynamics of the cross coupling of a dual drive gantry can be modeled via the flexibility of ball bearings of linear guides and the rotation of the crossbeam.

IV. SYNCHRONIZATION CONTROL SCHEMES

In this section, two synchronization controllers will be presented: a synchronization controller with cross-coupled compensation and a synchronization controller with thrust allocation. The former is used as a typical example of existing synchronization controls for comparison, while the latter is the proposed one with internal force regulation.

A. Synchronization Control With Cross-Coupled Compensation

1) **Synchronization Error Definition:** As illustrated in Fig. 5, with Y_1 and Y_2 axes representing positions of the two drives, a perfect motion synchronization of the two drives is represented by the straight line contour of $\gamma = 45^\circ$, i.e., $y_1 = y_2$. Thus, motion synchronization of the two drives becomes the same as the straight-line contour following in Fig. 5, which can be solved using the cross-coupled coordinated contouring control

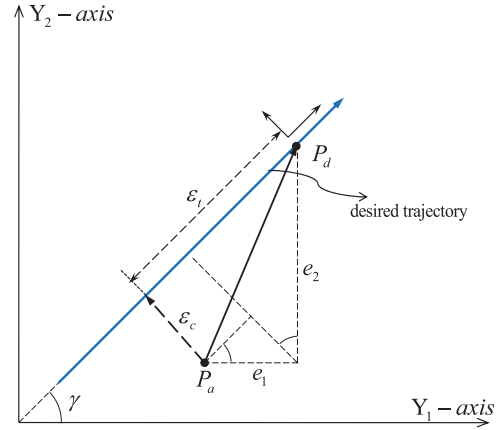


Fig. 5. Definition of synchronization error.

by treating the motion synchronization error as the contouring error in [19].

Let P_d be the desired position command on the desired contour in Fig. 5 that corresponds to the desired motion of the dual drive gantry $y_d(t)$, i.e., $P_d = \mathbf{y}_d(t) = [y_d(t), y_d(t)]^T$, and P_a the actual positions of the dual drives at any time, i.e., $P_a = \mathbf{y}(t) = [y_1(t), y_2(t)]^T$. Motion tracking errors of the two axes are thus represented by the position tracking errors $P_d P_a$ of $\mathbf{e} = [e_1, e_2]^T = \mathbf{y}(t) - \mathbf{y}_d(t)$. Motion synchronization error or contour error ϵ_c can then be calculated by the distance from P_a to the desired contour as

$$\epsilon_c = -\sin(\gamma) \cdot e_1 + \cos(\gamma) \cdot e_2. \quad (31)$$

Defining projection of the position tracking errors onto the desired contour as $\epsilon_t = \cos(\gamma) \cdot e_1 + \sin(\gamma) \cdot e_2$, we have the following coordinates transformation

$$\boldsymbol{\epsilon} = \mathbf{T}\mathbf{e}, \quad \mathbf{T} = \begin{bmatrix} -\sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \quad (32)$$

where $\boldsymbol{\epsilon} = [\epsilon_c, \epsilon_t]^T$.

2) **Error Dynamics:** As in [19], dynamics of the dual motors without considering the interaction forces by the crossbeam connection can be written as²

$$\mathbf{M}\ddot{\mathbf{y}} + \mathbf{B}\dot{\mathbf{y}} + \mathbf{A}\mathbf{S}_f(\dot{\mathbf{y}}) = \mathbf{u} + \mathbf{d}_n + \tilde{\mathbf{d}} \quad (33)$$

where $\mathbf{M} = \text{diag}[M_{n1}, M_{n2}]$, $\mathbf{B} = \text{diag}[B_{n1}, B_{n2}]$, and $\mathbf{A} = \text{diag}[A_{n1}, A_{n2}]$ represent the inertia matrix, the damping

²All terms are normalized with respect to the units of the current input commands for simplicity

$$\begin{aligned} G_{12c}(s) = & \frac{6.881 \times 10^4 (s^2 + 70.93s + 2.515 \times 10^5) (s^2 + 82.93s + 6.678 \times 10^5)}{(s^2 + 60.88s + 2.63 \times 10^5) (s^2 + 59.97s + 6.319 \times 10^5)} \\ & \times \frac{(s^2 + 73.08s + 9.083 \times 10^5) (s^2 + 992.1s + 1.999 \times 10^6)}{(s + 802.4) (s^2 + 44.14s + 8.582 \times 10^5) (s^2 + 51.48s + 1.031 \times 10^6)} \\ & \times \frac{(s^2 + 62.1s + 2.329 \times 10^6) (s + 1.048 \times 10^4)}{(s^2 + 306.9s + 1.983 \times 10^6) (s^2 + 73.09s + 2.189 \times 10^6)}. \end{aligned} \quad (30)$$

matrix, and the matrix of Coulomb friction coefficients of the two linear motor drives, respectively, $\mathbf{S}_f(\dot{\mathbf{y}}) = [S_f(\dot{y}_1), S_f(\dot{y}_2)]^T$, $\mathbf{u} = [u_1, u_2]^T$, $\mathbf{d}_n + \tilde{\mathbf{d}}$ represent the vector of lumped modeling errors and external disturbances, in which $\mathbf{d}_n = [d_{n1}, d_{n2}]^T$ represents the vector of constant nominal values.

With transformation (32), the following error dynamics are obtained

$$\mathbf{M}_t \ddot{\mathbf{e}} + \mathbf{B}_t \dot{\mathbf{e}} + \mathbf{M}_y \ddot{\mathbf{y}}_d + \mathbf{B}_y \dot{\mathbf{y}}_d + \mathbf{A}_y \mathbf{S}_f(\dot{\mathbf{y}}) = \mathbf{u}_t + \mathbf{d}_t + \tilde{\mathbf{d}}_t \quad (34)$$

where $\mathbf{M}_t = \mathbf{TMT}^{-1}$, $\mathbf{B}_t = \mathbf{TBT}^{-1}$, $\mathbf{M}_y = \mathbf{TM}$, $\mathbf{B}_y = \mathbf{TB}$, $\mathbf{A}_y = \mathbf{TA}$, $\mathbf{u}_t = \mathbf{Tu}$, $\mathbf{d}_t = \mathbf{Td}_n$, $\tilde{\mathbf{d}}_t = \mathbf{T}\tilde{\mathbf{d}}$ are the terms after transformation and are linear in terms of a set of parameters such as inertia, damping, friction coefficients, and nominal disturbances, which are captured by the eight-dimensional vector of unknown parameters as $\theta_c = [M_{n1}, M_{n2}, B_{n1}, B_{n2}, A_{n1}, A_{n2}, d_{n1}, d_{n2}]^T$.

3) Adaptive Robust Controller: Defining a switching-function-like quantity $\mathbf{s} = \dot{\mathbf{e}} + \Lambda \mathbf{e}$, where $\Lambda = \text{diag}[\lambda_1, \lambda_2] > 0$, the following ARC law can be synthesized as in [19]:

$$\mathbf{u}_t = \mathbf{u}_a + \mathbf{u}_{s1} + \mathbf{u}_{s2} \quad (35)$$

in which $\mathbf{u}_a = -\Psi(\mathbf{y}, \dot{\mathbf{y}}, t) \hat{\theta}_c$ is the adaptive model compensation with $\Psi = [\Psi_1 \ \Psi_2]^T$ being a 2×8 matrix of known functions and $\hat{\theta}_c$ updated through a discontinuous projection type parameter adaption law of $\hat{\theta}_c = \text{Proj}_{\hat{\theta}_c}(\Gamma_c \tau_c)$, $\Gamma_c > 0$, $\tau_c = \Psi^T(\mathbf{y}, \dot{\mathbf{y}}, t) \mathbf{s}$ [1], [19].

$\mathbf{u}_{s1} = -\mathbf{K}_c \mathbf{s}$ with $\mathbf{K}_c$ being a symmetric positive definite matrix is used to stabilize the nominal system, and $\mathbf{u}_{s2}$ is a nonlinear robust feedback to attenuate the effect of various model uncertainties for a guaranteed robust performance [19].

B. Synchronization Control With Thrust Allocation

1) Nonlinear Dynamics for Controller Design: For controller design, nonlinear frictions and uncertainties are added back to the Y -axis motion dynamics in (7) as

$$M \ddot{y}_G = F_{m1} + F_{m2} - F_{r1} - F_{r2} + d_y \quad (36)$$

where F_{m1} , F_{m2} , F_{r1} and F_{r2} are given by (3) and d_y represents all other lumped uncertainties and disturbances. For simplicity, $F_{r1}(\dot{y}_1)$ and $F_{r2}(\dot{y}_2)$ will be approximated by $F'_{r1}(\dot{y}_G) = B_1 \dot{y}_G + A_1 S_f(\dot{y}_G)$ and $F'_{r2}(\dot{y}_G) = B_2 \dot{y}_G + A_2 S_f(\dot{y}_G)$. Equation (36) can then be simplified to

$$M_k \ddot{y}_G = u_1 + k_m u_2 - B_k \dot{y}_G - A_k S_f(\dot{y}_G) + d_k \quad (37)$$

where $M_k = M/K_1$, $k_m = K_2/K_1$, $B_k = (B_1 + B_2)/K_1$, $A_k = (A_1 + A_2)/K_1$, and $d_k = d/K_1$ with $d = d_y + F'_{r1} + F'_{r2} - F_{r1} - F_{r2}$. Let Δ_n be a constant representing the unknown dc component of d_k and denote a vector of four unknown parameters as

$$\theta = [\theta_1, \dots, \theta_4]^T = [M_k, B_k, A_k, \Delta_n]^T. \quad (38)$$

The nonlinear dynamic equation (37) is then linearly parameterized by θ as

$$M_k \ddot{y}_G = v - B_k \dot{y}_G - A_k S_f(\dot{y}_G) + \Delta_n + \tilde{\Delta} \quad (39)$$

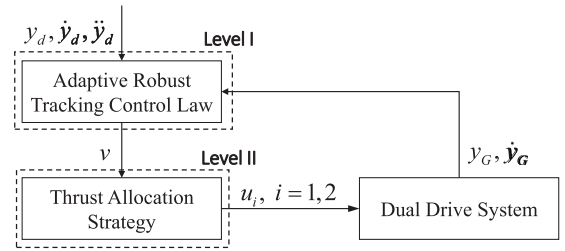


Fig. 6. Proposed synchronization control structure.

where $v = u_1 + k_m u_2$ is the virtual control input representing the total thrust generated and $\tilde{\Delta} = d_k - \Delta_n$ represents the time-varying portion of the lumped modeling uncertainties d_k .

2) Overall Control Structure: To avoid excessive internal forces for the smooth operation of dual drive systems, it is necessary to keep the steady-state rotational angle at zero or very small. Noting the rotational dynamics (8) and the fact that the total moment $\sum_{i=1}^2 M_{\alpha i}$ in (9) is a proportional feedback of the rotational angle, such a steady-state requirement can be satisfied if $(F_{m2} - F_{r2})l_2 - (F_{m1} - F_{r1})l_1$ is kept at zero or very small. With this in mind, the overall synchronization control structure shown in Fig. 6 is proposed. Namely, in level I, ARC theory is applied to synthesize the total thrust control input v for the motion dynamics (39) such that the position of the inertia load $y_G(t)$ will track its desired trajectory $y_d(t)$ with a guaranteed transient and steady-state performance even in the presence of parametric uncertainties in θ and uncertain nonlinearities $\tilde{\Delta}$; the details are given in Section IV-B3. Subsequently, in level II, the actual control inputs to the two linear motors u_1 and u_2 are determined based on the total thrust control input v synthesized in level I [i.e., (48)] and a thrust allocation strategy detailed in Section IV-B4 to keep $(F_{m2} - F_{r2})l_2 - (F_{m1} - F_{r1})l_1$ zero or small to avoid large internal forces.

3) Adaptive Robust Motion Tracking Controller: It is assumed that the extents of parametric uncertainties and uncertain nonlinearities are known, i.e.,

$$\theta \in \Omega_\theta \triangleq \{\theta : \theta_{\min} \leq \theta \leq \theta_{\max}\} \quad (40)$$

$$\tilde{\Delta} \in \Omega_\Delta \triangleq \{\tilde{\Delta} : |\tilde{\Delta}| \leq \delta_\Delta\} \quad (41)$$

where $\theta_{\min} = [\theta_{1\min}, \dots, \theta_{4\min}]^T$, $\theta_{\max} = [\theta_{1\max}, \dots, \theta_{4\max}]^T$ and δ_Δ are known. Let $\hat{\theta}$ be the estimate of θ and $\tilde{\theta}$ denote the estimation error (i.e., $\tilde{\theta} = \hat{\theta} - \theta$). In view of (40), the following discontinuous projection based adaptation law can be used

$$\dot{\hat{\theta}} = \text{Proj}_{\hat{\theta}}(\Gamma \tau) \quad (42)$$

where $\text{Proj}_{\hat{\theta}}(\bullet)$ is the projection mapping defined in the same way as in [4], $\Gamma > 0$ is a diagonal matrix, and τ is an adaptation function to be synthesized later.

Let $e_y = y_G - y_d$ denote the output tracking error. Define a switching-function-like quantity as

$$s = \dot{e}_y + k_1 e_y = \dot{y}_G - \dot{y}_{eq}, \quad \dot{y}_{eq} = \dot{y}_d - k_1 e_y \quad (43)$$

where $k_1 > 0$ is a positive gain. If s is small or converges to zero, then the tracking error e_y will be small or converge to zero since the transfer function from s to e_y is stable. So the rest of

the design is to make s as small as possible. Define a positive semidefinite function

$$V(t) = \frac{1}{2} M_k s^2. \quad (44)$$

Differentiating V and noting (39)

$$\dot{V} = s[v - M_k \ddot{y}_{eq} - B_k \dot{y}_G - A_k S_f(\dot{y}_G) + \Delta_n + \tilde{\Delta}] \quad (45)$$

where $\ddot{y}_{eq} = \ddot{y}_d - k_1 \dot{e}_y$. The terms in (45) are linearly parameterized as

$$M_k \ddot{y}_{eq} + B_k \dot{y}_G + A_k S_f(\dot{y}_G) - \Delta_n = -\varphi^T \theta \quad (46)$$

where $\varphi = [-\ddot{y}_{eq}, -\dot{y}_G, -S_f(\dot{y}_G), 1]^T$ is known as the regressor. Equation (45) can thus be rewritten as

$$\dot{V} = s(v + \varphi^T \theta + \tilde{\Delta}). \quad (47)$$

Noting the structure of (47), the following ARC law is proposed:

$$v = v_a + v_s; \quad v_a = -\varphi^T \hat{\theta} \quad (48)$$

in which v_a is the adjustable model compensation needed for perfect tracking and v_s is a robust control law to be synthesized later. Substituting (48) into (47)

$$\dot{V} = s(v_s - \varphi^T \tilde{\theta} + \tilde{\Delta}). \quad (49)$$

The robust control law v_s consists of the following two terms:

$$v_s = v_{s1} + v_{s2}; \quad v_{s1} = -k_2 s \quad (50)$$

where v_{s1} is used to stabilize the nominal system, which is chosen to be a simple proportional feedback with k_2 being a positive constant. And v_{s2} is a nonlinear feedback used to attenuate the effect of model uncertainties for a guaranteed robust performance, which is synthesized to satisfy the following two conditions:

$$\begin{aligned} i \quad & s\{v_{s2} - \varphi^T \tilde{\theta} + \tilde{\Delta}\} \leq \eta \\ ii \quad & s v_{s2} \leq 0 \end{aligned} \quad (51)$$

where η is a design parameter which can be arbitrarily small. One smooth example function of v_{s2} satisfying (51) is given by $v_{s2} = -\frac{1}{4\eta} h^2 s$, where h is any smooth function satisfying $h \geq \|\theta_{\max} - \theta_{\min}\| \|\varphi\| + \delta_{\Delta}$. With the above ARC law, using the same method as in [22], the following theoretic results can be obtained:

Theorem 1: With the adaptation function in (42) chosen as

$$\tau = \varphi s \quad (52)$$

the ARC control law (48) guarantees that

a) In general, all signals are bounded. Furthermore, the positive semidefinite function V defined by (44) is bounded above by

$$V(t) \leq \exp(-\lambda t) V(0) + \frac{\eta}{\lambda} [1 - \exp(-\lambda t)] \quad (53)$$

where $\lambda = 2k_2/\theta_{1\max}$.

b) If after a finite time t_0 , there exist parametric uncertainties only (i.e., $\tilde{\Delta} = 0, \forall t \geq t_0$), then, in addition to results in A, zero steady-state tracking error is also achieved, i.e., $e_y \rightarrow 0$ and $s \rightarrow 0$ as $t \rightarrow \infty$.

4) Thrust Allocation: To avoid large internal forces, the control inputs of two motors will be generated by the following thrust allocation strategy:

$$u_1 + k_m u_2 = v \quad (54)$$

$$K_1 l_1 u_1 - K_2 l_2 u_2 - \hat{F}_{r1} l_1 + \hat{F}_{r2} l_2 = 0 \quad (55)$$

where v in (54) is the virtual control input synthesized by ARC law (48), and $\hat{F}_{r1}$ and $\hat{F}_{r2}$ are the estimated frictions of the two linear guides. Noting (8), the condition (55) is to regulate the internal forces due to the physical coupling of the two axes to a level around zero. As frictions in most applications are relatively much smaller than the electromagnetic forces generated to drive the stage, the condition (55) could be simply implemented by letting $K_1 l_1 u_1 - K_2 l_2 u_2 = 0$, i.e., $u_1/u_2 = k_m l_2/l_1$.

V. COMPARATIVE EXPERIMENTAL RESULTS

Comparative control experiments are carried out on the dual drive gantry stage described in Section III with a sampling period of $T_s = 0.2$ ms.

A. Experiment Set1

To investigate the effect of physical coupling and bearings flexible mode on the achievable performance of traditional pure motion synchronization control strategy, experiments are first carried out using the compensation described in Section IV-A with four different sets of controller gains as follows. In the experiments, the smooth function $S_f(\bullet)$ is chosen as $\frac{2}{\pi} \arctan(9000\bullet)$. As in [19], the robust control part $\mathbf{u}_{s1} + \mathbf{u}_{s2}$ is chosen as $-\mathbf{K}_r \mathbf{s}$ with a large enough feedback gain $\mathbf{K}_r$ in implementation. The adaptation rates are set as $\Gamma_c = \text{diag}[20, 20, 20, 20, 10, 10, 3000, 3000]$. The initial parameter estimates are set at $\hat{\theta}_c(0) = [0.57, 0.3, 0.23, 0.79, 0.15, 0.27, 0, 0]^T$. The following sets of controller gains are compared when tracking a desired sinusoidal trajectory with an amplitude of 0.15 m and frequency of 0.5 Hz:

C1a: $\Lambda = \text{diag}[80, 60]$ and $\mathbf{K}_r = \text{diag}[50, 35]$;

C1b: $\Lambda = \text{diag}[120, 80]$ and $\mathbf{K}_r = \text{diag}[70, 50]$;

C1c: $\Lambda = \text{diag}[150, 100]$ and $\mathbf{K}_r = \text{diag}[90, 60]$;

C1d: $\Lambda = \text{diag}[200, 120]$ and $\mathbf{K}_r = \text{diag}[120, 70]$.

The experimental results in terms of tracking errors are shown in Fig. 7. In general, increasing the controller gains leads to an improved motion tracking performance and decreasing of synchronization errors, which are the situations seen from C1a to C1c. However, due to the appearance of the dominant flexible mode around 300 rad/s, too large feedback gains could easily cause control input chattering. In fact, with the gains in C1d, the control input chatting problem becomes so severe that worst tracking performance and largest synchronization errors exhibit in Fig. 7 instead.

B. Experiment Set2

Experiments are then carried out to compare the above traditional synchronization control with the proposed one presented in Section IV:

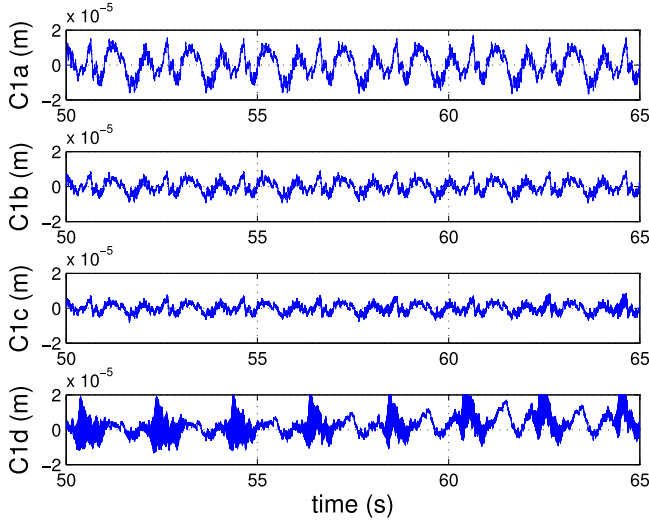


Fig. 7. Comparison of tracking errors for Set1.

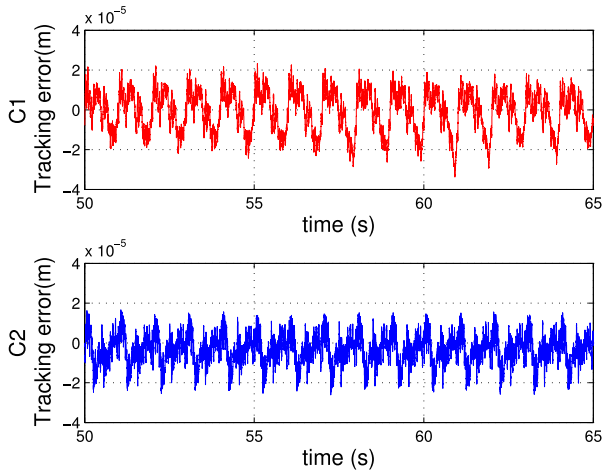


Fig. 8. Comparison of tracking errors for Set2.

C1: The *Synchronization Control With Cross-Coupled Compensation* described in Section IV-A with $\Lambda = \text{diag}[120, 80]$ and $\mathbf{K}_r = \text{diag}[70, 50]$;

C2: The proposed *Synchronization Control With Thrust Allocation* in Section IV. Theoretically, $v_{s2} = -k_{r2}(y, \dot{y}, t)s$ should be a nonlinear feedback as given in $v_{s2} = -\frac{1}{4\eta}h^2s$ to satisfy the robust performance requirement (51) globally. As in [19], a sufficiently large constant gain k_{r2} is used instead for simplicity. With this simplification, the robust control law in (50) is implemented as $v_s = -k_r s$, where k_r represents the summation of k_2 and k_{r2} . The following controller parameters are used: $k_1 = 120$, $k_r = 70$, and $\Gamma = \text{diag}[20, 20, 10, 3000]$ with the initial parameter estimates of $\hat{\theta}(0) = [0.85, 0.88, 0.37, 0]^T$.

The desired trajectory is a sinusoidal trajectory with amplitude of 0.15 m and frequency of 1 Hz. Experimental results of the synchronization errors of two linear motors are shown in Fig. 8. In addition, the root-mean-square value of the tracking errors, $\|e_y\|_2 = (\frac{1}{T} \int_0^T |e_y|^2 dt)^{1/2}$, is used to measure the

TABLE I
PERFORMANCE AND CONTROL INPUT INDEXES, Set2

Algorithms	$\ e_y\ _2 (\mu m)$	$\ e_y\ _\infty (\mu m)$	$\ u\ _{rms} (V)$
C1	10.05	33.71	4.98
C2	10.08	31.56	4.18

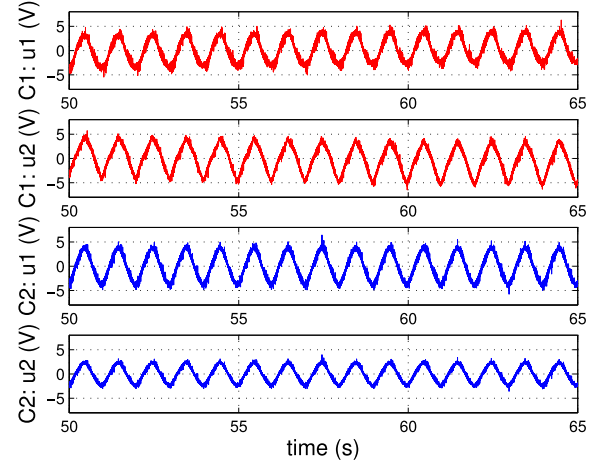


Fig. 9. Comparison of control inputs for Set2.

average tracking performance, and the maximum absolute value of the tracking error, $\|e_y\|_\infty = \max_t \{e_y\}$, is used as an index for the transient performance. The indexes for both algorithms are shown in Table I. It is seen from these results that the two controllers have similar motion tracking performance.

The control inputs of the two motors for both control schemes are shown in Fig. 9. In addition, the average control inputs of two motors, $\|u\|_{rms} = (\frac{1}{T} \int_0^T (|u_1| + |u_2|)^2 dt)^{1/2}$, is used as an index for the extent of control efforts and the corresponding experimental results are given in Table I. Due to the strongly coupled dynamics between the two drive axes, “pull and drag” phenomena could exhibit even when motion tracking performance looks good and extra control efforts may be needed to deal with the resistance caused by the internal forces. Noting the plots in Fig. 9 and the control effort index in Table I, the effectiveness of the proposed control scheme lies in the fact that it achieves the same level of motion performance but with a smaller control effort.

C. Experiment Set3

Comparative experiments are then performed with the same desired trajectory as in Set2 but with larger controller gains as follows:

C3: The *Synchronization Control With Cross-Coupled Compensation* with $\Lambda = \text{diag}[180, 150]$ and $\mathbf{K}_r = \text{diag}[110, 90]$;

C4: The proposed *Synchronization Control With Thrust Allocation* with $k_1 = 180$ and $k_r = 110$.

The results of the tracking errors and the control inputs of two motors for both control schemes are plotted in Figs. 10 and 11, respectively. It is seen that C3 does not perform well in such a situation due to the excitation of the neglected dominant flexible mode. The two motor drives interfere with each other severely,

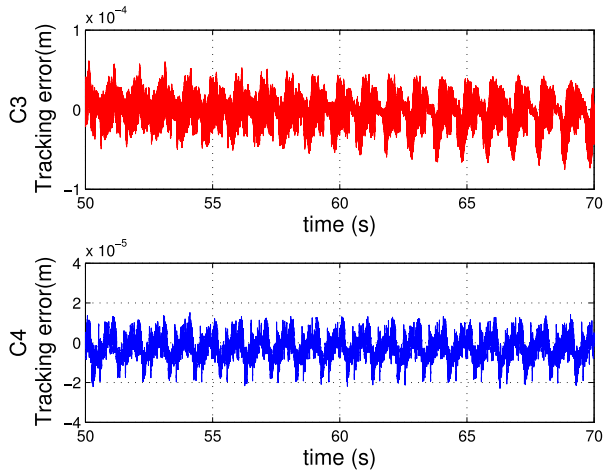


Fig. 10. Comparison of tracking errors for Set3.

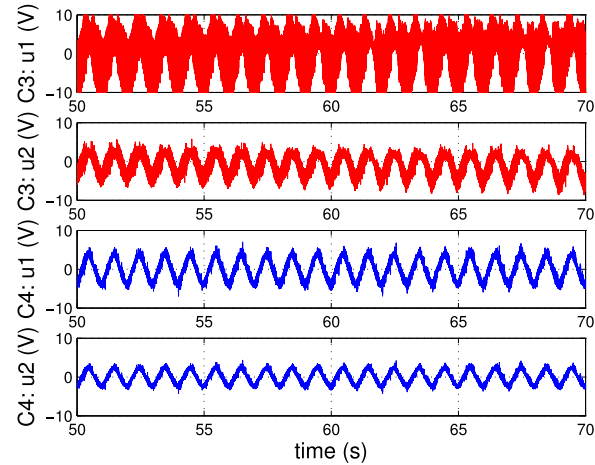


Fig. 11. Comparison of control inputs for Set3.

TABLE II
PERFORMANCE AND CONTROL INPUT INDEXES, SET3

Algorithms	$\ e_y\ _2 (\mu m)$	$\ e_y\ _\infty (\mu m)$	$\ u\ _{rms} (V)$
C3	20.14	75.03	7.46
C4	6.78	22.92	4.21

leading to the control input saturation of motor 1 and the system almost went unstable. In comparison, the proposed strategy, the controller C4, does not have such interfering problems and achieves even smaller motion tracking errors than in *Set2*. These claims are also verified by the quantitative results in terms of indexes shown in Table II.

In summary, the proposed synchronization controller achieves better motion tracking performance with lesser control efforts in practice. The thrust allocation proposed in this paper is simple and yet effective in regulating the internal forces and eliminating the severe interference of the two motion drives.

D. Experiment Set4

To verify the robustness of the proposed synchronization controller with thrust allocation, a 9.8-kg payload is mounted on the gantry to study the effect of parameter variations on the

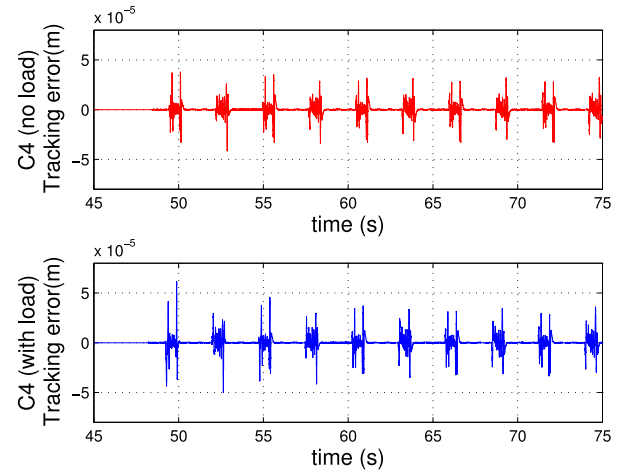


Fig. 12. Comparison of transient performance for Set4.

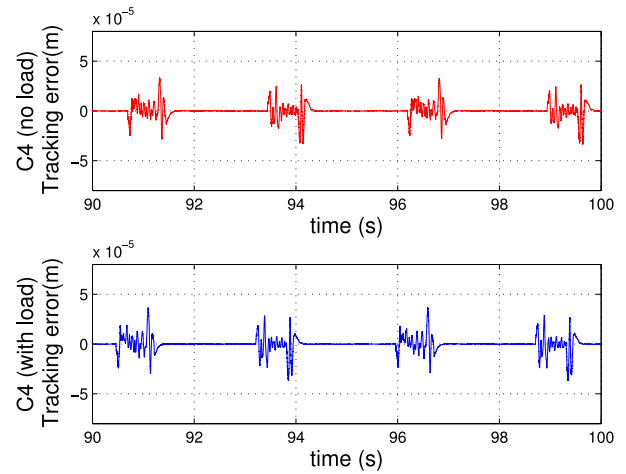


Fig. 13. Comparison of tracking errors for Set4 (after a finite time).

same controller as C4 in Set3. The desired trajectory is a point-to-point S-curve with a travel displacement of 0.3 m, a maximum velocity of 0.5 m/s and a maximum acceleration of 5 m/s². Fig. 12 shows the comparison of transient performance. It is seen that in the case with a payload on the gantry, due to the much bigger deviation of the initial inertia parameter estimation error, the maximum error during the first several periods is much larger (up to 61.47 μm) than that without load (41.39 μm). However, the controller still works well under load variation during the steady-state as shown in Fig. 13; it achieves almost the same level of tracking performance as no-load case despite the change of inertia load.

Figs. 14 and 15 show the time history of parameter estimates of C4 in both cases. As seen from the results, due to the adaptive learning process of the proposed controller, the estimations of inertia load converge to reasonable ranges in both cases. That also explains why the tracking performances become better after several finite periods in Fig. 12. Even though some of the other parameter estimates such as the damping and friction coefficients do not converge, good tracking performances have been obtained in implementation since the proposed ARC controller

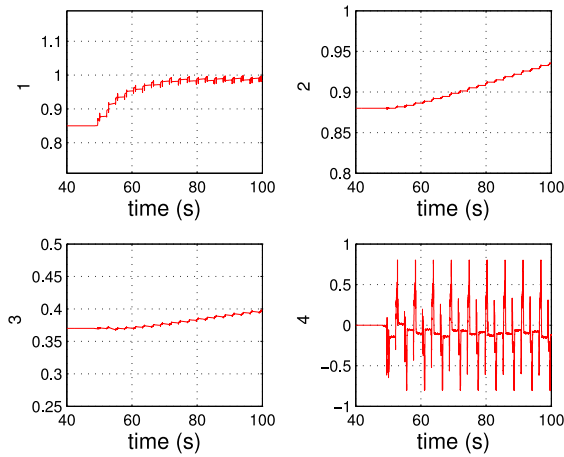


Fig. 14. Estimation of parameters for Set4 (no load).

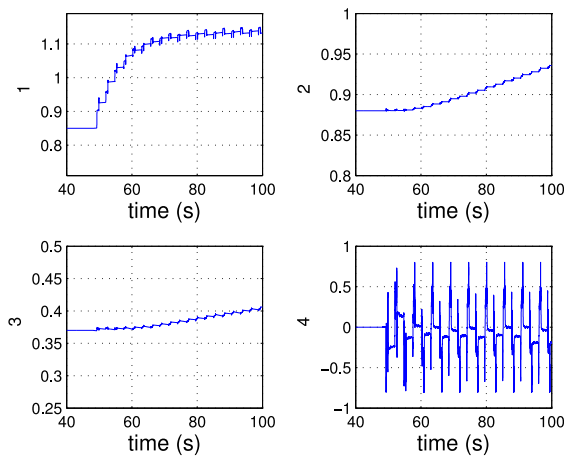


Fig. 15. Estimation of parameters for Set4 (with load).

focuses on the total adjustable model compensation for perfect tracking instead of the convergence of all parameter estimates.

VI. CONCLUSION

In this paper, a novel dynamic model based on distributed lumped bearing stiffness of linear guides is derived for a dual drive gantry system. The model is then validated through extensive frequency response identification experiments. The results reveal that the flexibility of ball bearings is the cause of the dominant flexible mode and the small rotation of the crossbeam due to motion synchronization errors of two linear drives could lead to significantly large interaction forces between dual axes. A novel synchronization controller with thrust allocation is subsequently presented, which not only synchronizes motions of the two motors for better overall motion tracking performance but also avoids exciting the dominant flexible mode through regulating the internal forces between the two axes. The proposed synchronization strategy is also compared with the existing coordinated control based ones. Comparative experimental results verify the effectiveness and superiority of the proposed synchronization controller in practical applications as well. Some future works include in-depth examination of

the dominant flexible mode to investigate possible mechanical redesigns to alleviate its effect and/or to actively control such modes for further control performance improvement.

REFERENCES

- [1] L. Xu and B. Yao, "Adaptive robust precision motion control of linear motors with negligible electrical dynamics: Theory and experiments," *IEEE/ASME Trans. Mechatronics*, vol. 6, no. 4, pp. 444–452, Dec. 2001.
- [2] Y. Hong and B. Yao, "A globally stable high performance adaptive robust control algorithm with input saturation for precision motion control of linear motor drive system," *IEEE/ASME Trans. Mechatronics*, vol. 12, no. 2, pp. 198–207, Apr. 2007.
- [3] J. Wu, Z. Xiong, K.-M. Lee, and H. Ding, "High-acceleration precision point-to-point motion control with look-ahead properties," *IEEE Trans. Ind. Electron.*, vol. 58, no. 9, pp. 4343–4352, Sep. 2011.
- [4] B. Yao, C. Hu, L. Lu, and Q. Wang, "Adaptive robust precision motion control of a high-speed industrial gantry with cogging force compensations," *IEEE Trans. Control Syst. Technol.*, vol. 19, no. 5, pp. 1149–1159, Sep. 2011.
- [5] L. Lu and B. Yao, "A performance oriented multi-loop constrained adaptive robust tracking control of one-degree-of-freedom mechanical systems: Theory and experiments," *Automatica*, vol. 50, no. 4, pp. 1143–1150, 2014.
- [6] K. Cho, J. Kim, S. B. Choi, and S. Oh, "A high-precision motion control based on a periodic adaptive disturbance observer in a PMLSM," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 5, pp. 2158–2171, Oct. 2015.
- [7] Z. Chen, B. Yao, and Q. Wang, " μ -synthesis based adaptive robust control of linear motor driven stages with high-frequency dynamics: A case study," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 3, pp. 1482–1490, Jun. 2015.
- [8] H.-H. Chiang, K.-C. Hsu, and I.-H. Li, "Optimized adaptive motion control through an SoPC implementation for linear induction motor drives," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 1, pp. 348–360, Feb. 2015.
- [9] J. Zheng, H. Wang, Z. Man, J. Jin, and M. Fu, "Robust motion control of a linear motor positioner using fast nonsingular terminal sliding mode," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 4, pp. 1743–1752, Aug. 2015.
- [10] C. Li, B. Yao, X. Zhu, and Q. Wang, "Dual drive system modeling and analysis for synchronous control of an h-type gantry," in *Proc. IEEE/ASME Conf. Adv. Intell. Mechatronics*, Busan, Korea, Jul. 2015, pp. 214–219.
- [11] R. D. Lorenz and P. B. Schmidt, "Synchronized motion control for process automation," in *Proc. IEEE Industry Appl. Soc. Annu. Meeting*, 1989, pp. 1693–1698.
- [12] B. Chu, S.-S. Kim, D. Hong, H.-K. Park, and J.-M. Park, "Optimal cross-coupled synchronizing control of dual-drive gantry system for a SMD assembly machine," *JSME Int. J. Ser. C Mech. Syst., Mach. Elements Manuf.*, vol. 47, no. 3, pp. 939–945, 2004.
- [13] M.-C. Tsai and B.-H. Shen, "Synchronisation control of parallel dual inverted pendulums driven by linear servomotors," *IET Control Theory Appl.*, vol. 1, pp. 320–327, 2007.
- [14] K.-K. Tan, S.-Y. Lim, S. Huang, H.-F. Dou, and T.-S. Giam, "Coordinated motion control of moving gantry stages for precision applications based on an observer-augmented composite controller," *IEEE Trans. Control Syst. Technol.*, vol. 12, no. 6, pp. 984–991, Nov. 2004.
- [15] D. Sun, X. Shao, and G. Feng, "A model-free cross-coupled control for position synchronization of multi-axis motions: Theory and experiments," *IEEE Trans. Control Syst. Technol.*, vol. 15, no. 2, pp. 306–314, Mar. 2007.
- [16] Y. Koren, "Cross-coupled biaxial computer control for manufacturing systems," *ASME J. Dyn. Syst., Meas., Control*, vol. 102, pp. 265–272, 1980.
- [17] G. T.-C. Chiu and M. Tomizuka, "Contouring control of machine tool feed drive systems: A task coordinate frame approach," *IEEE Trans. Control Syst. Technol.*, vol. 9, no. 1, pp. 130–139, Jan. 2001.
- [18] S.-S. Yeh and P.-L. Hsu, "Estimation of the contouring error vector for the cross-coupled control design," *IEEE/ASME Trans. Mechatronics*, vol. 7, no. 1, pp. 44–51, Mar. 2002.
- [19] C. Hu, B. Yao, and Q. Wang, "Coordinated adaptive robust contouring controller design for an industrial biaxial precision gantry," *IEEE/ASME Trans. Mechatronics*, vol. 15, no. 5, pp. 728–735, Oct. 2010.
- [20] C. Hu, B. Yao, and Q. Wang, "Integrated direct/indirect adaptive robust contouring control of a biaxial gantry with accurate parameter estimations," *Automatica*, vol. 46, no. 4, pp. 701–707, Apr. 2010.

- [21] C. Hu, B. Yao, and Q. Wang, "Global task coordinate frame based contouring control of linear-motor-driven biaxial systems with accurate parameter estimations," *IEEE Trans. Ind. Electron.*, vol. 58, no. 11, pp. 5195–5205, Nov. 2011.
- [22] B. Yao, C. Hu, and Q. Wang, "An orthogonal global task coordinate frame for contouring control of biaxial systems," *IEEE/ASME Trans. Mechatronics*, vol. 17, no. 4, pp. 622–634, Aug. 2012.
- [23] C. Hu, B. Yao, and Q. Wang, "Adaptive robust repetitive control of an industrial biaxial precision gantry for contouring tasks," *IEEE Trans. Control Syst. Technol.*, vol. 19, no. 6, pp. 1559–1568, Nov. 2011.
- [24] Y. Xiao and K. Zhu, "Optimal synchronization control of high-precision motion systems," *IEEE Trans. Ind. Electron.*, vol. 53, no. 4, pp. 1160–1169, Jun. 2006.
- [25] I. Garciaherrerros, X. Kestelyn, J. Gomand, and P. J. Barre, "Decoupling basis control of dual-drive gantry stages for path-tracking applications," in *Proc. IEEE Int. Symp. Ind. Electron.*, Jul. 2010, pp. 131–136.
- [26] F.-J. Lin, P.-H. Chou, C.-S. Chen, and Y.-S. Lin, "DSP-based cross-coupled synchronous control for dual linear motors via intelligent complementary sliding mode control," *IEEE Trans. Ind. Electron.*, vol. 59, no. 2, pp. 1061–1073, Feb. 2012.
- [27] H.-K. Park, S.-S. Kim, J.-M. Park, T.-Y. Cho, and D. Hong, "Dynamics of dual-drive servo mechanism," in *Proc. IEEE Int. Symp. Ind. Electron.*, 2001, pp. 1996–2000.
- [28] C.-S. Teo, K.-K. Tan, S.-Y. Lim, S. Huang, and E.-B. Tay, "Dynamic modeling and adaptive control of a H-type gantry stage," *Mechatronics*, vol. 17, no. 7, pp. 361–367, 2007.
- [29] H. Ohta and E. Hayashi, "Vibration of linear guideway type recirculating linear ball bearings," *J. Sound Vib.*, vol. 235, no. 5, pp. 847–861, 2000.



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